

# Statistical & Numerical Methods HW2

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*Technion*

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# 1 Area probability with an angle

A needle of length  $l$  is dropped on a grid of parallel lines with distance  $a$  between them, such that  $l < a$ .

1. What is the chance the needle hits the net if it is dropped such that it must be perpendicular to the net?
2. What is the chance it hits the net if it falls at a random angle?

## 1.1 Perpendicular needle

In this case it's simply the length of the needle divided by the size of the gap

$$P = \frac{l}{a} \quad (1)$$

## 1.2 at angle

In this case there actually needs to be a calculation. First we notice that given we already know the angle at which the needle dropped, the chance is simply

$$P(hit) = \frac{l}{a} \sin \theta \quad (2)$$

therefore

$$P(hit) = \int_0^\pi d\theta P(hit|\Theta = \theta)P(\Theta = \theta) \quad (3)$$

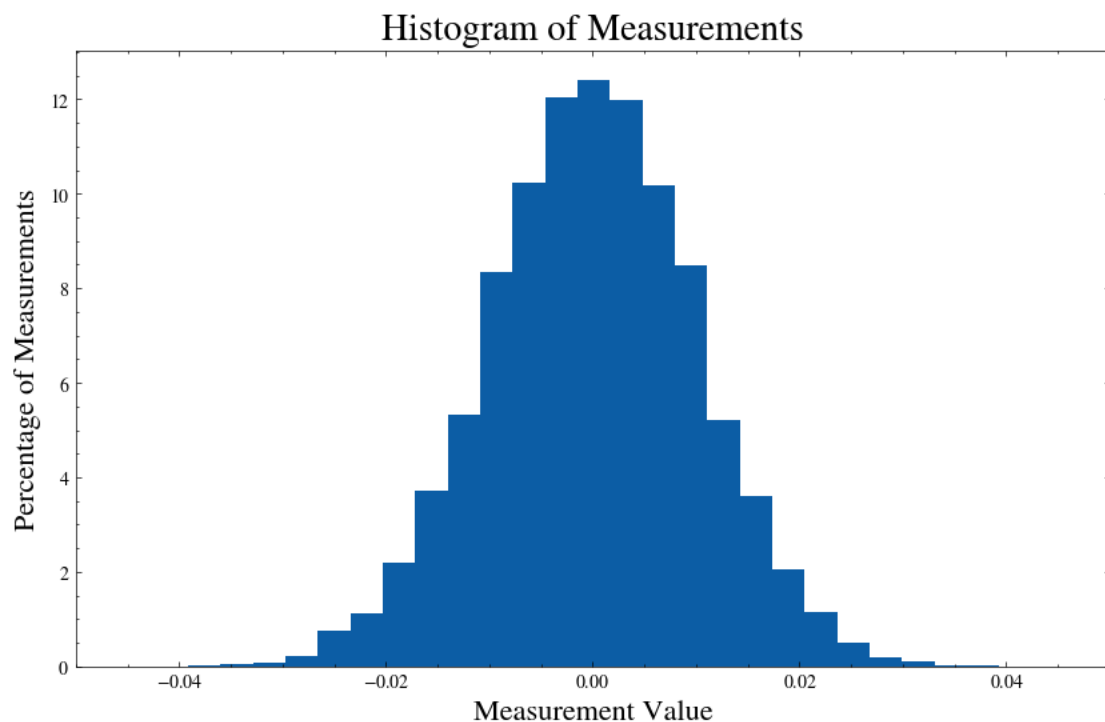
$$= \frac{l}{a} \int_0^\pi d\theta \sin \theta P(\Theta = \theta) \quad (4)$$

$$= \frac{2l}{\pi a} \quad (5)$$

## 2 Probability Distribution

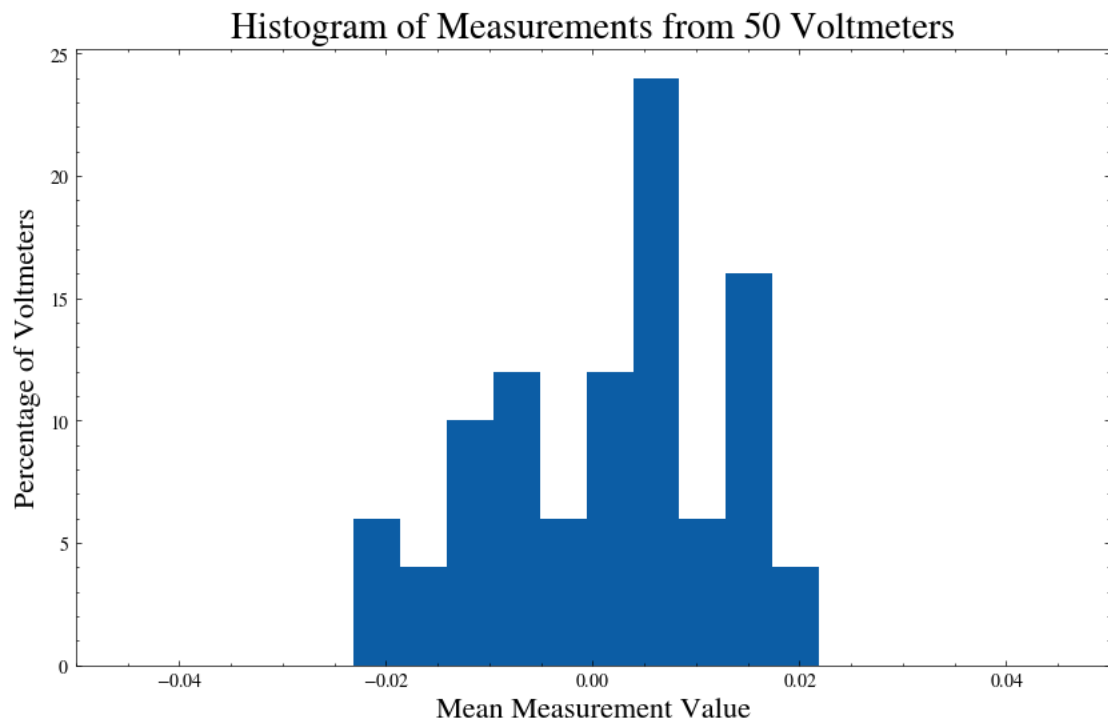
1. Create and display a gaussian distribution of 10,000 measurements of 0 volt with a single voltmeter and a standard deviation of 0.01.
2. Show a histogram of 50 measurements for a measurement of 0 volt which is performed with 50 different measurement devices.
3. Show a combined measurement of 0 volt performed as follows: 20 measurements with a device whose standard deviation is 0.1, 10 with a device with a deviation of 0.05, 5 with an std to 0.01, and 15 with an std of 0.02.
4. We'll use the measurement device from part 1 and measure the following set of voltages, each voltage measured 5 times: 0.5v, 1.0v, 3.9v, 7.8v, 10v, 13.4v, 15.2v, 17.5v, 21.1v, 25v. Show the results as a scatter plot (the wanted voltage as a function of the measured voltage).
5. Performed a linear fit for the measurements in part 4 and use the tools you know to estimate how good the fit is.

### 2.1 0 volts



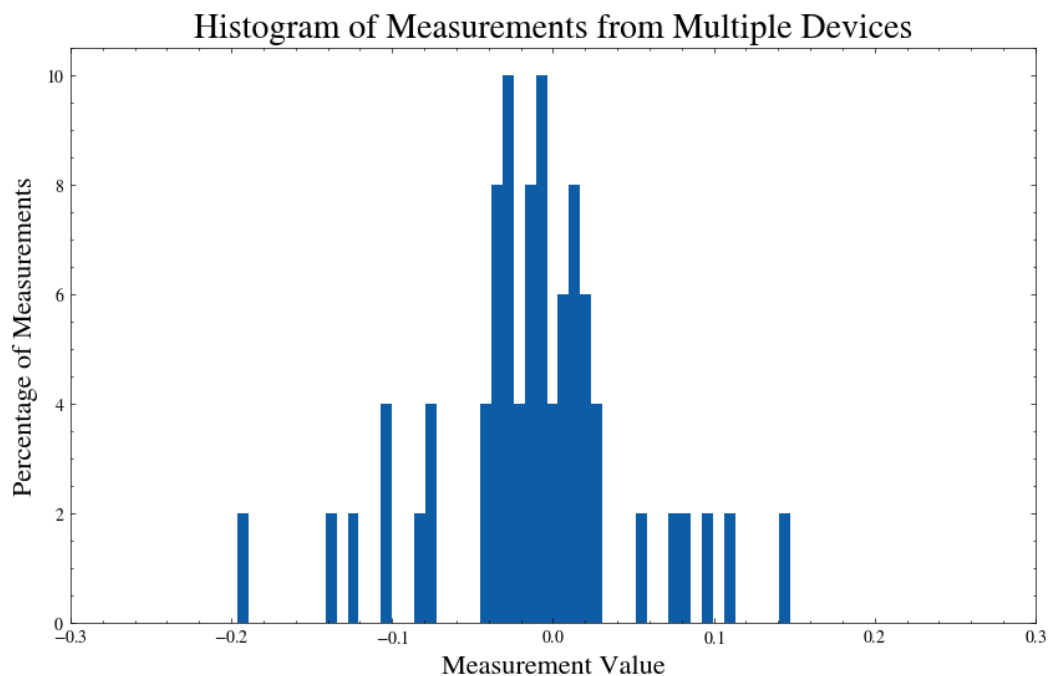
**Figure 1** – Histogram of 10,000 measurements from a gaussian distribution with  $\mu = 0$  and  $\sigma = 0.01$ .

## 2.2 50 measurements



**Figure 2** – Histogram of 50 measurements using 50 different gaussian distributions with  $\mu = 0$  and  $\sigma = 0.01$ .

## 2.3 composite measurement

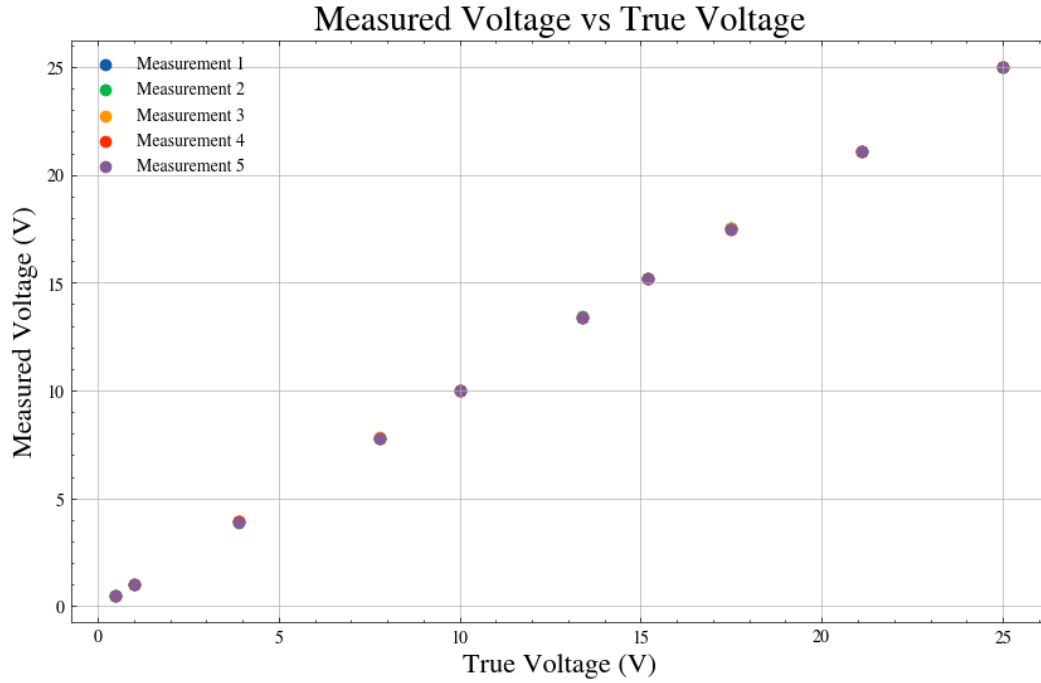


**Figure 3** – Histogram of measurements of 0V, where 20 have  $\sigma = 0.1$ , 10 have  $\sigma = 0.05$ , 5 have  $\sigma = 0.01$ , and 15 have  $\sigma = 0.02$ .

Using the calculations for the weighted mean and uncertainty on it we find

$$\bar{x} = 0.004854 \qquad \sigma_{\bar{x}} = 0.003270 \qquad (6)$$

## 2.4 Multiple Voltages



**Figure 4** – Histogram of measurements of several voltages, each measured 5 times, all with the same device.

## 2.5 Linear Fit

Performing a linear fit we find

$$y = (0.9999 \pm 0.0002) x + (0.0019 \pm 0.0002) \qquad R^2 = 0.999 \qquad (7)$$

### 3 Gravity measurements

1. How many cycles will be required to measure  $g$  within 1%?
2. How many measurements will be needed to verify the theory within 1.65 sigma?
3. Calculate the error for  $g$  and write down an estimator for its error
4. Estimate the average of gravity using the measurements, then estimate the variance using the measurements. Compare to the theoretical.
5. Calculate  $\hat{\sigma}_\sigma$  for parts 3 and 4, what causes the difference between these values?

#### 3.1 g within 1 percent

First we reverse the relation

$$T = 2\pi\sqrt{\frac{l}{g}} \implies g = \frac{4\pi^2 l}{T^2} \quad (8)$$

therefore the error in  $g$  is

$$\Delta g = 4\pi^2 l \sqrt{\frac{4}{T^6}} \Delta T^2 = \frac{8\pi^2 l}{T^3} \Delta T \quad (9)$$

The number of cycles is such that

$$T = \frac{t}{N} \implies \Delta T = \frac{\Delta t}{N} \quad (10)$$

returning, we want

$$\frac{\Delta g}{g} = \frac{\frac{8\pi^2 l}{T^3} \Delta T}{\frac{4\pi^2 l}{T^2}} = \frac{2}{T} \Delta T \leq 0.01 \quad (11)$$

inserting the cycles

$$\frac{2\Delta t}{t} \leq 0.01 \quad (12)$$

inserting  $\Delta t = 0.5$

$$\frac{1\text{ s}}{t} \leq 0.01 \implies t \geq 100\text{ s} \quad (13)$$

the cycle is of length

$$T = 2\pi\sqrt{\frac{l}{g}} \approx 2\text{ s} \quad (14)$$

therefore

$$N = \frac{t}{T} \geq \frac{100\text{ s}}{2\text{ s}} = 50 \quad (15)$$

#### 3.2 1.65 sigma

The gravitational acceleration is

$$g = \frac{GM}{r^2} \quad (16)$$

so

$$g_{\text{sea}} = \frac{GM}{r_{\text{sea}}^2} \qquad g_{\text{everest}} = \frac{GM}{r_{\text{everest}}^2} \quad (17)$$

we can express the everest acceleration using the sea acceleration

$$g_{\text{everest}} = g_{\text{sea}} \left( \frac{r_{\text{sea}}}{r_{\text{everest}}} \right)^2 \quad (18)$$

we can insert the known numbers

$$g_{\text{everest}} = 9.8 \text{ m/s}^2 \cdot \left( \frac{6400}{6409} \right)^2 = 9.7725 \quad (19)$$

we want

$$\frac{g_{\text{sea}} - g_{\text{everest}}}{\Delta g} = \frac{0.0275}{\Delta g} > 1.65 \implies \Delta g < 0.01667 \text{ m/s}^2 \quad (20)$$

but

$$\Delta g = \frac{8\pi^2 l}{T^3} \Delta T = \frac{8\pi^2 l}{T^3} \frac{\Delta t}{N} \quad (21)$$

rearranging

$$N = \frac{8\pi^2 l}{T^3} \frac{\Delta t}{\Delta g} \quad (22)$$

inserting all the numbers we find

$$N = \frac{8\pi^2 \cdot 1 \text{ m}}{(2 \text{ s})^3} \frac{0.5 \text{ s}}{0.01667 \text{ m/s}^2} > 296 \quad (23)$$

### 3.3 Student Experiment

This time both the time and length have errors, therefore

$$\Delta g = \sqrt{\left( \frac{\partial g}{\partial l} \Delta l \right)^2 + \left( \frac{\partial g}{\partial T} \Delta T \right)^2} \quad (24)$$

where

$$g = \frac{4\pi^2 l}{T^2} \quad (25)$$

giving us

$$\Delta g = \sqrt{\left( \frac{4\pi^2}{T^2} \cdot 0.07 \text{ m} \right)^2 + \left( \frac{8\pi^2 l}{T^3} \cdot 0.05 \text{ s} \right)^2} \approx 0.85 \text{ m/s}^2 \quad (26)$$

but since we have 50 measurements the total standard deviation is

$$\sigma = \frac{\Delta g}{\sqrt{50}} = 0.12 \text{ m/s}^2 \quad (27)$$

we can use the formula for an estimator for a variance with a known mean:

$$\hat{\Delta g} = \sqrt{\frac{1}{N} \sum_i \left( g_i - \frac{4\pi^2 l_i}{T_i^2} \right)^2} \quad (28)$$

### 3.4 With Measurements

A good estimate for the mean is simply the average of the measurements:

$$\hat{g} = \bar{g} = 9.792 \text{ m/s}^2 \quad (29)$$

and the variance can be calculated using the formula for an unknown mean

$$\hat{V} = \frac{1}{N-1} \sum_i (g_i - \bar{g})^2 = 0.094 \text{ (m/s}^2\text{)}^2 \quad (30)$$

therefore

$$\sigma = 0.307 \text{ m/s}^2 \quad (31)$$

### 3.5 sigma sigma

$$\hat{\sigma}_{\sigma,c} = \frac{\sigma}{\sqrt{2N}} = 0.012 \text{ m/s}^2 \quad (32)$$

$$\hat{\sigma}_{\sigma,d} = \frac{\sigma}{\sqrt{2N}} = 0.0307 \text{ m/s}^2 \quad (33)$$

The sigma for section 3 is smaller than that of section 4. This is because in section 3 we took the value of  $g$  as an assumption for a value we know, while in section 4 we did not. This prior knowledge changes our uncertainty.



## 4 "Good" Estimators

$\hat{a}, \hat{b}, \hat{c}, \hat{d}$  are unbiased estimators such that  $\langle a \rangle = \langle b \rangle = \mu$ ;  $\langle c \rangle = 2 \langle d \rangle = \lambda$  with variances  $\sigma_{\hat{a}} = \sigma_{\hat{b}} = 4\sigma_{\hat{c}} = 2\sigma_{\hat{d}}$

1. Check whether the following estimators are unbiased, and fix any that are so they are unbiased

$$\frac{\hat{c} + 2\hat{d}}{5} \quad e^{c+d} \quad \frac{\hat{a}}{\hat{b}^2} \quad \sum_{i=1}^{\infty} \frac{\hat{b}^{2i-1}}{(2i-1)!} \quad (34)$$

2. Which of these estimators is most efficient?

### 4.1 bias check

$$\left\langle \frac{c + 2d}{5} \right\rangle = \frac{\langle c \rangle + 2 \langle d \rangle}{5} = \frac{2}{5} \lambda \quad (35)$$

there it is biased. We can fix it by making the estimator

$$\frac{5 \hat{c} + 2\hat{d}}{2} \quad (36)$$

onto the next one

$$\langle e^{c+d} \rangle = e^{1.5\lambda} \quad (37)$$

so it is also biased, and the fix is

$$\log_{1.5e} (e^{c+d}) \quad (38)$$

next

$$\left\langle \frac{\hat{a}}{\hat{b}^2} \right\rangle = \frac{\mu}{\mu^2} = \frac{1}{\mu} \quad (39)$$

this is also biased, the fix is for example

$$\frac{\hat{a}^3}{\hat{b}^2} \quad (40)$$

next

$$\sum_{i=1}^{\infty} \frac{\hat{b}^{2i-1}}{(2i-1)!} \quad (41)$$

is the Taylor series for sinh so

$$\left\langle \sum_{i=1}^{\infty} \frac{\hat{b}^{2i-1}}{(2i-1)!} \right\rangle = \langle \sinh(\hat{b}) \rangle = \sinh(\langle \hat{b} \rangle) = \sinh \mu \quad (42)$$

which is biased and the fix is

$$\sinh^{-1} \left( \sum_{i=1}^{\infty} \frac{\hat{b}^{2i-1}}{(2i-1)!} \right) \quad (43)$$

## 4.2 most efficient

The most efficient estimator is the one with the lowest variance. We'll calculate all the variances and express them using  $\sigma_a$  for convenience.

$$\hat{V}\left(\frac{5}{2}\frac{\hat{c}+2\hat{d}}{5}\right) = \hat{V}\left(\frac{\hat{c}}{2} + \hat{d}\right) = \frac{\sigma_c^2}{4} + \frac{25}{4}\sigma_c^2 = 6.5\sigma_c^2 = \frac{25}{20}\sigma_a^2 \quad (44)$$

next

$$\hat{V}\left(\frac{\hat{a}^3}{\hat{b}^2}\right) = \frac{\sigma_a^6}{\sigma_b^4} = \sigma_a^2 \quad (45)$$

next

$$\hat{V}(\sinh^{-1}(\sinh(b))) = \hat{V}(b) = \sigma_b^2 = \sigma_a^2 \quad (46)$$

and

$$\hat{V}(\log_{1.5e}(e^{c+d})) = \hat{V}(c+d) = \sigma_c^2 + \sigma_d^2 = \frac{1}{25}\sigma_a^2 + \frac{1}{4}\sigma_a^2 = \frac{29}{100}\sigma_a^2 \quad (47)$$

so the most efficient estimator is the last one.